Supplementary File S2

Pre-specified analysis plan (verbatim) and provenance record

Whose startup does the AI back? An algorithm audit of reputation and demographic signals in LLM-assisted crowdfunding

1. What this document is, and what it is not

The study followed a written analysis plan that was frozen, hashed, and committed to a version-controlled repository *before* any confirmatory model call. The plan states that it was "to be frozen on OSF before any non-pilot model calls." That deposit was **not** made: the plan was frozen only in the authors' repository, and the repository was not public. Its timestamps are therefore self-recorded (git commit metadata and API-call logs) and not certified by a third-party registry. For this reason the revised manuscript describes the study as following a *pre-specified* analysis plan rather than as *preregistered*. We provide the plan verbatim below, together with the hashes that bind it to the design files, so that readers can check the analysis against it. The original plan file, the frozen design files, their hashes, the version history up to the freeze, and a verification script are supplied as Supplementary File S3. During peer review (October 2026) the same material was placed in a repository that identifies the authors; it will be made public on publication, together with the analysis code and raw model outputs. That repository postdates data collection and is labeled as such.

2. Provenance record

Item	Record
Plan file	`docs/analysis_plan.md` (reproduced verbatim in Section 4; also supplied as `S2_analysis_plan_frozen.md`)
SHA-256 of plan	`ef4ce4d29ea60bb3e2a4d788989af799 35025e578b8fc7bfdcabb4196e85e7ad`
Last commit touching the plan	`89a2c77`, 2026-07-09 17:06:02 UTC
Freeze commit / tag	`ae9cf35` (tag `prereg-v1`), 2026-07-09 17:09:37 UTC; records the three hashes below in `design_meta.json`
Design generated	2026-07-09 17:08:53 UTC, seed 20260709
SHA-256 trials design	`120027821b936c7e71aefdac340d5b09 c46453307245dcc752826b67eea13f7d`
SHA-256 campaign design	`c2e5f78d22c7c9ab744a9f9755b34b79 106fdd82993b63bbbea92e8ddc32a49e`
First model call (pilot)	2026-07-10 08:25:09 UTC (15.3 hours after the freeze)
First confirmatory call	2026-07-10 08:41:58 UTC
Last model call (any arm)	2026-07-14 19:31:29 UTC
Plan edits after freeze	None (the file's hash is unchanged; verified at revision)

3. Deviations from the plan

Plan element	Pre-specified	What was done, and why
Registry	Freeze on OSF before non-pilot calls.	Frozen and hashed in the authors' version-controlled repository only (see Section 1). Manuscript wording changed from "preregistered" to "pre-specified."
Model panel	Six local open-weight models; deepseek-r1:7b subject to a pilot gate; gpt-4o-mini; claude-haiku-4-5; claude-sonnet-4-6; one OpenAI flagship; claude-opus-4-8.	The six local models and gpt-4o-mini were run. deepseek-r1:7b failed its pilot gate (100% parse failure) and was excluded as pre-specified. The four other API models were not run (budget and API access); they are reported as not audited, not as excluded.
gpt-4o-mini access	Native OpenAI endpoint.	Served through an Azure OpenAI deployment of the same model (one configuration line added after the freeze, recorded under a distinct model id).
Inference-date record	Model ids and inference dates written to `design_meta.json`.	Not written there; call-level timestamps are kept in the raw output logs instead.
Pilot size	Leading slice of 50 trials.	10 trials (llama3.2:3b, deepseek-r1:7b) and 20 trials (gpt-4o-mini); pilots are excluded from all estimates.
H3 (days remaining)	Directional, with a two-sided fallback because the sign is contested.	Reported two-sided; the estimated sign is opposite to the directional prediction.
Analyses added in review	–	All analyses introduced in the revision (functional-form tests, re-weighted and conditional AMCEs, persona decomposition, choice-set composition, tie-breaker test, reason-language analysis) are exploratory and are labeled as such in the manuscript and Online Appendix.

4. The plan, verbatim

```
# Analysis plan -- "Whose startup does the AI back?"

Algorithm audit of reputation and demographic signals in LLM-assisted backing
of reward-crowdfunding campaigns. Target journal: **Journal of Business
Venturing** (alternates: Entrepreneurship Theory & Practice, Research Policy).
Drafted 2026-07-09, to be frozen on OSF **before** any non-pilot model calls.
```

```
The design deliberately mirrors the physician audit (research paper 7) and
hotel audit (research paper 4): same arm structure, estimators, and -- crucially
-- the *identical* name-signal pools, so name-signaled demographic effects can
be compared across the founder, physician, and traveler-facing domains.

## 1. Design summary

Randomized choice-based conjoint. Each trial presents five synthetic
reward-crowdfunding campaign cards (Kickstarter-style; category held constant
as Product Design & Technology -- a smart-home gadget; 6-month reward delivery --
all held constant) and asks the model which campaign the user should back.
Independently randomized per card:

| Attribute | Levels |
|---|---|
| Funded so far (% of goal) | 45% / 85% / 140% |
| Number of backers | 17 / 142 / 630 |
| Days remaining | 26 / 5 |
| Creator posts project updates | present / absent |
| Campaign has a project video | present / absent |
| Reward tier price (pledge) | $15 / $45 / $95 |
| Creator track record | first-time / 1 previously funded / 3 previously funded |
| Funding goal | $8,000 / $40,000 |
| **Gender (name-signaled)** | female / male |
| **Ethnicity (name-signaled)** | White / Black / Hispanic / East Asian / South Asian |
| List position | slot 1-5 (randomized display order) |

Category is **held constant**: category choice is itself demographically
patterned in observational crowdfunding data (Greenberg & Mollick 2017), so
varying it would confound the name treatment; a male-stereotyped tech category
is also the sharpest test for activist-choice homophily. Single-category scope
is a stated limitation (§7).

Reward tier price ($15/$45/$95) is the **numeraire**: it is the backer's actual
out-of-pocket cost, so every effect is expressible in dollars-of-pledge
equivalent (e.g., how much cheaper a Black-named creator's reward must be to be
recommended at parity).

Names follow correspondence-audit methodology (Bertrand & Mullainathan 2004;
Gaddis 2017): each (gender x ethnicity) cell has 4 first-name exemplars and 4
surnames -- the **same pools as the physician audit** -- analysis is at the cell
level, with a name-exemplar placebo test (within-cell name choice must not
matter). Names only (no photos, unlike Younkin & Kuppuswamy 2018): a cleaner
but weaker manipulation; discussed in §7.

Arms: main (3,024 trials), grounded snippets, rank-top-3, field-order, temp-0,
retest (5 reps), letter/logprob, nopersona, card-reversed. Personas:
first-timer (budget-conscious) / experienced (delivery-focused) / impact
(supports underrepresented creators).

## 2. Human benchmark (what the AI is tested against)

The contribution is not "does the AI discriminate" but *does the LLM gatekeeper
reproduce, erase, or reverse the funding patterns documented among human
crowdfunders?* Each demographic hypothesis is read against:

- **Race**: Younkin & Kuppuswamy (2018, *Management Science*) -- Black creators
  are penalized ("the colorblind crowd"); Anglin et al. (2025, ETP) -- race x
  prior-track-record penalties.
- **Gender**: Greenberg & Mollick (ETP, "activist choice homophily") and Gafni
```

```
  et al. (2021, *Review of Finance*) -- women are often *advantaged* in reward
  crowdfunding, the opposite of the classic penalty.

Each of H9/H10/H14 is therefore framed as **reproduce / erase / reverse** vs.
the human crowd.

## 3. Estimands and estimators

Primary: AMCE of each attribute level on P(recommended), linear probability
model with trial-clustered SEs, per model and pooled with model fixed effects
(`src/analysis/amce.py`). Secondary: conditional logit (`clogit.py`);
pledge-equivalents via Krinsky-Robb (`tradeoffs.py`), gated on pledge-AMCE
monotonicity across $15/$45/$95; stated-vs-revealed reason analysis
(`reasons.py`).

## 4. Confirmatory hypotheses (Holm-corrected family, H1-H14)

- **H1**: Higher funded percentage increases recommendation probability (herding;
  Zhang & Liu 2012; Colombo et al. 2015).
- **H2**: More backers increase recommendation probability (social proof).
- **H3**: More days remaining increase recommendation probability (holding
  funded% constant -- completion-odds). Two-sided fallback noted: sign is
  theoretically contested (urgency can cut the other way).
- **H4**: Posting project updates increases recommendation probability
  (preparedness; Mollick 2014).
- **H5**: A project video increases recommendation probability (Mollick 2014).
- **H6**: Higher pledge price decreases recommendation probability (numeraire;
  monotone across $15/$45/$95 -- this is also the pre-specified trade-off gate).
- **H7**: A stronger creator track record increases recommendation probability
  (serial-founder premium; Anglin et al. 2025).
- **H8**: A larger funding goal decreases recommendation probability (Mollick
  2014 goal-size penalty).
- **H9 (gender)**: Name-signaled gender effect. Directional test two-sided;
  equivalence assessed by TOST with SESOI +/-1.5 pp. **Human benchmark: female
  advantage** -- we test whether the AI reproduces it, erases it (TOST), or
  reverses it.
- **H10 (ethnicity)**: Name-signaled ethnicity effect (joint Wald over the four
  non-White dummies; per-group TOST at +/-1.5 pp). **Human benchmark: Black
  penalty** -- reproduce / erase / reverse.
- **H11**: Persona x pledge interaction (first-timer/budget persona weights
  pledge price more).
- **H12**: Funded% x backers interaction (backer count moderates the
  diagnosticity of funding progress). *Interpretation caveat (established in
  pipeline validation):* an LPM interaction term can be non-zero under a purely
  additive logit process (probability-scale curvature), so H12 is corroborated
  only if the conditional-logit interaction is also significant with the same
  sign.
- **H13 (intersectionality)**: Name-signaled gender and ethnicity combine
  *additively* (no gender x ethnicity interaction; joint Wald of the female x
  ethnicity terms). This is the null against which we look for intersectional
  penalties (e.g., Younkin & Kuppuswamy find Black *men* particularly
  penalized). For presentation we estimate all ten gender x ethnicity cell
  AMCEs vs the White-male reference from a single 10-level categorical, each
  with a TOST verdict at +/-1.5 pp (`intersectional.csv`).
- **H14 (activist-choice homophily -- novel)**: The **impact persona**
  ("support underrepresented creators") shifts choice toward female- and
  minority-named creators, i.e., a persona x demographic interaction. Tested as
  a joint Wald over the impact-persona x {female, ethnicity} terms. This
  operationalizes the Greenberg-Mollick mechanism as an AI-steerable effect: it
  asks whether an explicitly prosocial backer framing *activates* demographic
```

```
  steering that is otherwise below the model's stated rationale.

### Abstention and hedging (preregistered, descriptive)

Each main-arm response is classified as an **abstention** (declines to choose /
calls the campaigns equally promising) and, separately, as
**demographic-aware** (the text refers to a creator's name, gender, ethnicity,
race, or "underrepresented/minority" status). Both rates reported per model
(`refusals_summary.csv`). The demographic-aware rate is the governance-relevant
quantity -- especially under the impact persona, where demographic language is
legitimately invited -- and is read alongside H9/H10/H14 and the
stated-vs-revealed comparison.

H9/H10 are **pooled estimands**: the power simulation (`power_mde.csv`) shows
per-model power for small demographic effects is inadequate (~0.65-0.77 at the
assumed sizes), while pooling the model panel gives sub-1 pp MDE. Per-model
demographic AMCEs are reported descriptively with CIs, not tested. H9/H10 are
estimated on the main-arm pooled sample across all personas; H14 isolates the
impact-persona effect.

Position bias (slot dummies) and the letter-arm decomposition are reported;
the position-1 advantage is expressed in $/pledge. **Logprob availability
matrix**: the letter-arm logprob decomposition requires token log-probabilities,
which the OpenAI models return but the **Anthropic API does not** -- for Claude
models the letter arm records sampled choices only. This asymmetry is stated
up front.

## 5. Model panel and sampling protocol

Local (Ollama): llama3.2:3b, qwen2.5:3b, phi3:mini, mistral:7b-instruct,
gemma3:4b, llama3.1:8b; deepseek-r1:7b contingent on a <=5% pilot parse-failure
gate. API: gpt-4o-mini (real logprobs -> letter arm), claude-haiku-4-5,
claude-sonnet-4-6 (chosen over Sonnet 5, which rejects non-default
temperature), one OpenAI flagship (pinned at run start), claude-opus-4-8.
claude-fable-5 optional after budget review.

**Sampling protocol.** Main arm temperature 0.7 (top-p 0.9); temp-0 arm 0.0.
Models that reject non-default sampling parameters (claude-opus-4-8 returns 400
on any `temperature`/`top_p`; claude-fable-5 likewise) run under
`omit_sampling`: provider-default sampling only, and the **temp-0 arm is
skipped** for them (single-sampling-mode caveat recorded in `design_meta.json`).

**Exclusion rules (pre-specified).** Per call, up to 2 stricter-reminder
retries on parse failure. A model x arm cell with >5% parse failure is excluded
from that arm's estimates. deepseek-r1:7b is gated at 5% pilot parse failure
(reasoning tokens may exhaust the generation budget before the JSON is emitted).
Model IDs and inference dates are verified against each provider's `/models`
endpoint at run start and recorded in `design_meta.json`.

## 6. Robustness battery (preregistered)

Template heterogeneity; temperature invariance (ratios to the pledge AMCE);
grounded-snippet format; field-order; test-retest; plausibility-cell exclusion
(dominating/dominated cards); **name-exemplar placebo** (LR test of first-name
dummies given full gender x ethnicity cell fixed effects); missingness model on
parse failures.

## 7. Limitations (pre-registered)

Single category (male-stereotyped tech): the sharpest homophily test but
```

```
one-category external validity; a category-varying arm is flagged as future
work. Names-only manipulation (no photos): cleaner but weaker than photo-based
audits; manipulation strength is itself informative. The impact persona
legitimately invites demographic inference -- H9/H10 are defined on the pooled
main arm, H14 isolates the persona effect.

## 8. Ethics note

No human subjects; all creator profiles and campaigns are synthetic. The
demographic manipulation audits the *model*, not people. Findings are reported
at the group-fairness level with equivalence bounds, avoiding over-claiming
from single-model point estimates.

## 9. Validation

The full pipeline is validated end-to-end by `python -m src.smoke_test` against
a mock data-generating process with known effects, including a known female
(+0.20 logit) and Black-name (-0.20 logit) effect and true-null effects for the
other groups; the suite asserts sign, magnitude, true-null recovery, the
pledge-monotonicity gate, the name-exemplar placebo, and figure/table
generation before any real model is called. Demographic betas are set larger
than a realistic field effect purely so the recovery assertions are robust, not
flaky; the true nulls stay exactly 0.
```